

Decision under Uncertainty and Portfolio Analysis

Lecture 1: IRFA and MMMEF

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Source: Slides can be downloaded from [xiangyuqu@weebly.com](https://xiangyuqu.weebly.com).

Risk and uncertainty

- ▶ X is outcome set.
- ▶ $\Delta(X) = \{p : X \rightarrow [0, 1] : \sum_{x \in X} p(x) = 1\}$ is the set of lotteries.
- ▶ S is state space.
- ▶ $F = \{f : S \rightarrow X\}$ is the set of acts.
- ▶ $\succsim$ is a binary relation.

Objective risk

Outcomes and their probabilities are specified: a lottery

$$P = (x_1, p_1; \dots; x_n, p_n), \quad \sum_i p_i = 1.$$

Subjective uncertainty

An act maps states into outcomes:

$$f : S \rightarrow X.$$

Beliefs about states need not be given in advance.

The benchmark theory separates **tastes over outcomes** from **beliefs about events**.

From expected value to expected utility

Expected value ranks lotteries by $\sum_i p_i x_i$. This can fail descriptively: sufficiently attractive rare prizes can make expected value explode, even when willingness to pay is modest.

$$V_{\text{EU}}(P) = \sum_{i=1}^n p_i u(x_i).$$

- ▶ u translates outcomes into utility.
- ▶ Probabilities enter linearly.
- ▶ Curvature of u captures attitude toward outcome risk.

In-class question

Why can a concave utility function give a finite valuation to a gamble with a very large or infinite expected monetary payoff?

Preferences over lotteries

Let $P \succsim Q$ mean that lottery P is weakly preferred to Q . For all lotteries P, Q, R ,

- ▶ **Completeness:** either $P \succsim Q$, or $Q \succsim P$, or both.
 - ▶ Any two lotteries can be compared.
- ▶ **Transitivity:** if $P \succsim Q$ and $Q \succsim R$, then $P \succsim R$.
 - ▶ Rankings are internally consistent.
- ▶ **Mixture Continuity:** if $P \succ Q$ and $Q \succ R$, then $Q \sim \alpha P + (1 - \alpha)R$ for some $\alpha \in (0, 1)$.
 - ▶ Small probability changes do not cause preference jumps.
- ▶ **Independence:** if $P \succsim Q$, then $\alpha P + (1 - \alpha)R \succsim \alpha Q + (1 - \alpha)R$ for all $\alpha \in (0, 1)$.
 - ▶ Mixing both options with the same third lottery preserves their ranking.

Theorem

A preference relation $\succsim$ on $\Delta(X)$ satisfies completeness, transitivity, mixture continuity and independence if and only if it is represented by an expected utility function $V_{\text{EU}}(P)$.

The independence axiom

For lotteries P, Q, R and $\alpha \in (0, 1)$:

$$P \succeq Q \iff \alpha P + (1 - \alpha)R \succeq \alpha Q + (1 - \alpha)R.$$

The common component R should not affect the comparison.

Operational interpretation

After a random device selects the common branch, the decision maker faces the same consequence under both options. Only the non-common branches differ.

In-class question

Does cancelling a common consequence remain compelling when that consequence is almost certain?

Expected utility over wealth

For random wealth $\tilde{x}$ with distribution F ,

$$V(F) = \mathbb{E}[U(\tilde{x})] = \int U(x) dF(x).$$

Assume $U'(x) > 0$, so more wealth is preferred.

First-order stochastic dominance

Moving probability from a lower outcome to a higher outcome raises expected utility for every increasing U .

Risk aversion

An individual is risk averse when, for every random wealth $\tilde{x}$,

$$U(\mathbb{E}[\tilde{x}]) \geq \mathbb{E}[U(\tilde{x})].$$

The certain mean is weakly preferred to the gamble.

Jensen's inequality

Risk aversion holds for every gamble if and only if U is concave. Strict concavity yields strict preference for every non-degenerate gamble.

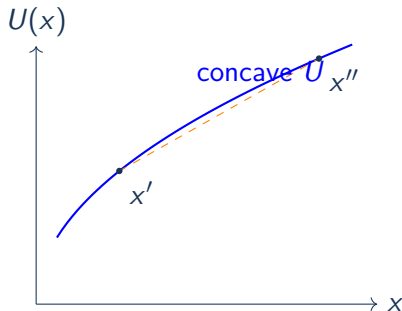
The geometry of concavity

Suppose $\tilde{x}$ equals x' with probability $2/3$ and x'' with probability $1/3$. Then

$$\bar{x} = \frac{2}{3}x' + \frac{1}{3}x'',$$

and concavity gives

$$U(\bar{x}) > \frac{2}{3}U(x') + \frac{1}{3}U(x'').$$



Certainty equivalent and risk premium

The certainty equivalent c solves

$$U(c) = \mathbb{E}[U(\tilde{x})].$$

The risk premium is

$$\pi(\tilde{x}) = \mathbb{E}[\tilde{x}] - c.$$

For a risk-averse person,

$$c \leq \mathbb{E}[\tilde{x}], \quad \pi(\tilde{x}) \geq 0.$$

In-class question

A gamble pays 0 or 100 with equal probability. What information beyond its mean determines the certainty equivalent?

Risk attitudes

Preference	Utility shape	Comparison with mean
Risk averse	Concave	$U(\mathbb{E}\tilde{x}) \geq \mathbb{E}U(\tilde{x})$
Risk neutral	Linear	$U(\mathbb{E}\tilde{x}) = \mathbb{E}U(\tilde{x})$
Risk loving	Convex	$U(\mathbb{E}\tilde{x}) \leq \mathbb{E}U(\tilde{x})$

These categories compare a gamble with its certain mean. They do not rank every pair of risky distributions.

Standard deviation as a measure of risk

$$\sigma^2 = \text{Var}(\tilde{x}) = \mathbb{E}[(\tilde{x} - \mathbb{E}\tilde{x})^2], \quad \sigma = \sqrt{\text{Var}(\tilde{x})}.$$

Mean-standard deviation analysis represents a prospect by (μ, σ) .

- ▶ It reduces a distribution to two numbers.
- ▶ Portfolio means and variances are easy to calculate.
- ▶ The reduction can discard information relevant to expected utility.

A riskless and a risky asset

Invest share a in a riskless asset with return r_0 and $1 - a$ in risky return $\tilde{r}$:

$$\tilde{r}_p = ar_0 + (1 - a)\tilde{r}.$$

Then

$$\mu_p = ar_0 + (1 - a)\mu_r, \quad \sigma_p = |1 - a|\sigma_r.$$

For $0 \leq a \leq 1$, attainable portfolios trace a line segment in (σ, μ) space. Borrowing at r_0 extends the line beyond the all-risky portfolio.

Two risky assets and diversification

For $\tilde{r}_p = a\tilde{r}_1 + (1 - a)\tilde{r}_2$,

$$\sigma_p^2 = a^2\sigma_1^2 + (1 - a)^2\sigma_2^2 + 2a(1 - a)\text{Cov}(\tilde{r}_1, \tilde{r}_2).$$

Covariance determines the shape of the attainable set in mean-standard deviation space.

Diversification

Low covariance can reduce portfolio standard deviation even when both assets are risky.

When two moments are sufficient

For quadratic utility $U(x) = \alpha x + \beta x^2$,

$$\begin{aligned}\mathbb{E}[U(\tilde{x})] &= \alpha \mathbb{E}[\tilde{x}] + \beta \mathbb{E}[\tilde{x}^2] \\ &= \alpha \mu + \beta(\mu^2 + \sigma^2).\end{aligned}$$

Expected utility depends only on mean and variance.

General result

Across unrestricted distributions, an expected-utility agent ranks all risks solely by mean and variance if and only if utility is quadratic.

Problems with quadratic utility

With $\beta < 0$, quadratic utility is concave, but

$$U'(x) = \alpha + 2\beta x$$

eventually becomes negative.

- ▶ Beyond the satiation point, more wealth lowers utility.
- ▶ Absolute risk aversion rises with wealth over the increasing region.
- ▶ Mean-variance analysis remains exact only under special utility or distributional restrictions.

Borch's critique

A ranking based only on (μ, σ) can declare two distributions indifferent even when one first-order stochastically dominates the other.

$$\tilde{x}_2 \succ_{FSD} \tilde{x}_1 \implies \mathbb{E}[U(\tilde{x}_2)] > \mathbb{E}[U(\tilde{x}_1)]$$

for every increasing U .

Lesson

Mean and standard deviation may discard distributional information that every monotone expected-utility agent values.

Entropy and payoff risk

For probabilities p_i , Shannon entropy is

$$H(P) = - \sum_i p_i \log p_i.$$

Entropy does not depend on the distance between outcomes. Therefore,

$$(49, 51; 0.5, 0.5) \quad \text{and} \quad (0, 100; 0.5, 0.5)$$

have identical entropy, although the second gamble has far greater payoff dispersion.

In-class question

What feature does entropy measure, and why does that differ from the economic cost of bearing payoff risk?

Risk as a partial order

The increasing-risk approach asks whether two equal-mean distributions satisfy a relation on which every risk-averse expected-utility agent agrees.

Partial ordering

Some equal-mean distributions remain incomparable because different concave utility functions rank them differently.

The relation does not rank every pair, but it is transitive.

Definition A: Risk is what all risk averters hate

F^* is riskier than F when they have the same mean and

$$\int U(x) dF^*(x) \leq \int U(x) dF(x)$$

for every concave utility function U .

This relation is called the concave order. With equal means, it also corresponds to second-order stochastic dominance in the relevant direction.

Definition B: Additional fair noise

Let

$$\tilde{y} = \tilde{x} + \tilde{\varepsilon}, \quad \mathbb{E}[\tilde{\varepsilon} \mid \tilde{x}] = 0.$$

The noise need not be independent of $\tilde{x}$. Its conditional variance may vary with $\tilde{x}$.

Conditional Jensen inequality

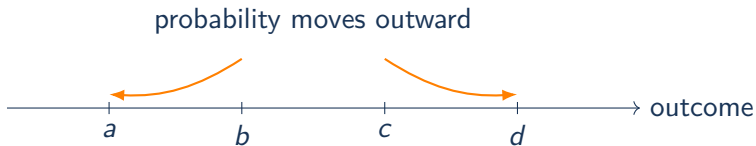
For concave U ,

$$\mathbb{E}[U(\tilde{y}) \mid \tilde{x}] \leq U(\tilde{x}) \implies \mathbb{E}U(\tilde{y}) \leq \mathbb{E}U(\tilde{x}).$$

F^* is riskier than F when they are the respective cumulative distribution functions of the random variables $\tilde{y}$ and $\tilde{x}$.

Definition C: mean-preserving spread

A mean-preserving spread moves probability from the center toward the tails while keeping the mean fixed.



F^* is riskier than F when F^* may be obtained from a sequence of mean preserving spread.

The integrated-CDF condition

For equal-mean distributions on $[0, M]$, F^* is riskier than F when

$$\int_0^x [F^*(t) - F(t)] dt \geq 0 \quad \forall x \in [0, M],$$

with equality at $x = M$.

- ▶ A single spread may produce one crossing of the CDFs.
- ▶ A sequence of spreads may produce several crossings.
- ▶ The integrated-CDF condition survives every sequence of spreads.

Rothschild-Stiglitz equivalence

For two equal-mean distributions, the following are equivalent:

1. every concave expected-utility agent prefers F to F^* ;
2. F^* equals F plus conditionally mean-zero noise;
3. mean-preserving spreads transform F into F^* ;
4. the integrated CDF of F^* lies weakly above that of F .

concave order $\iff$ fair noise $\iff$ mean-preserving spread

A discrete example

$$\tilde{x} = \begin{cases} 40 & \text{with probability } 1/2, \\ 60 & \text{with probability } 1/2, \end{cases} \quad \mathbb{E}\tilde{x} = 50,$$

while

$$\tilde{y} = \{20, 40, 60, 80 \text{ each with probability } 1/4. \quad \mathbb{E}\tilde{y} = 50.$$

$\tilde{y}$ is a mean-preserving spread of $\tilde{x}$. Every concave utility function weakly prefers $\tilde{x}$.

Variance versus increasing risk

If $\tilde{y} = \tilde{x} + \tilde{\varepsilon}$ and $\mathbb{E}[\tilde{\varepsilon} \mid \tilde{x}] = 0$, then

$$\text{Var}(\tilde{y}) = \text{Var}(\tilde{x}) + \mathbb{E}[\text{Var}(\tilde{\varepsilon} \mid \tilde{x})] \geq \text{Var}(\tilde{x}).$$

The converse generally fails.

Distinction

Higher variance alone does not imply that every risk-averse expected-utility agent dislikes the distribution. Concave order compares the whole distribution.

Comparative statics under increasing risk

Suppose the agent chooses a to maximize

$$\int U(x, a) dF(x; r),$$

where higher r makes $F(\cdot; r)$ riskier. The first-order condition is

$$\int U_a(x, a) dF(x; r) = 0.$$

If $U_a(x, a)$ is concave in x , an increase in risk lowers this expression at the original choice. With $U_{aa} < 0$, restoring the first-order condition requires a lower a .

The cross-derivative condition

Concavity of the marginal payoff from the choice means

$$U_{xxa}(x, a) \leq 0.$$

Under the regularity conditions in the article,

$$U_{xxa} < 0 \implies \text{a mean-preserving increase in risk lowers } a^*.$$

The sign reverses when $U_{xxa} > 0$.

In-class question

If a denotes investment in a risky activity, what could make the marginal payoff U_a concave in the uncertain return?

Comparison of the concepts

Concept	Question	Limitation
Concave utility	Is a gamble preferred to its mean?	Does not rank every pair of risks
Standard deviation	How dispersed are outcomes around the mean?	Ignores higher moments and shape
Increasing-risk order	Do all concave-utility agents agree?	Gives a partial ordering

Selected references

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-  Borch, K. (1969). A note on uncertainty and indifference curves. *Review of Economic Studies*.
-  Rothschild, M. and Stiglitz, J. (1970). Increasing risk I: A definition. *Journal of Economic Theory*.
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Relative risk attitude

Definition

u_1 is more risk-averse than u_2 if the former dislikes every lottery for which the latter is indifferent, at every wealth level:

$$\forall X, w_0 : \quad \mathbb{E}u_2(w_0 + X) = u_2(w_0) \implies \mathbb{E}u_1(w_0 + X) \leq u_1(w_0).$$

► Necessary and sufficient condition:

$$\exists \phi \text{ concave} : \quad u_1(w) = \phi(u_2(w)) \quad \forall w,$$

$$\begin{aligned} \mathbb{E}u_1(w_0 + X) &= \mathbb{E}\phi(u_2(w_0 + X)) && \text{definition of } \phi \\ &\leq \phi(\mathbb{E}u_2(w_0 + X)) && \text{Jensen} \\ &= \phi(u_2(w_0)) && u_2 \text{ indifferent} \\ &= u_1(w_0) && \text{definition of } \phi. \end{aligned}$$

Risk premium

- ▶ Maximum amount the agent is willing to pay to escape a zero-mean risk.
- ▶ Under risk aversion, the risk premium is nonnegative.
- ▶ It is defined by

$$\mathbb{E}u(w_0 + X) = u(w_0 - \pi).$$

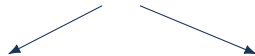
- ▶ More risk aversion implies a larger π .

Arrow–Pratt approximation

- Suppose that X is “small.”

$$A(w_0) = -\frac{u''(w_0)}{u'(w_0)} \quad (\text{absolute risk aversion}),$$

$$\pi(w_0, u, X) \approx \frac{1}{2} \text{Var}(X) A(w_0).$$



measure of the risk measure of risk aversion

Comparing risk aversion

For twice differentiable utility, Arrow–Pratt absolute risk aversion is

$$A(x) = -\frac{U''(x)}{U'(x)}.$$

Agent 1 is everywhere more risk averse than agent 2 when:

1. $A_1(x) \geq A_2(x)$ for all x ;
2. U_1 is an increasing concave transformation of U_2 ;
3. agent 1 rejects every gamble that agent 2 rejects at the same wealth.

This gives observable content to comparative curvature.

Decreasing absolute risk aversion (DARA)

- ▶ It is widely accepted that π decreases with initial wealth w_0 .
- ▶ This holds if and only if $A(w_0)$ decreases with w_0 .
- ▶ DARA is equivalent to

$$-\frac{u'''(z)}{u''(z)} \geq -\frac{u''(z)}{u'(z)} \quad \text{for all } z.$$

Some classical utility functions

- ▶ CARA: $u(z) = -\exp(-Az)$; $A(z) = A$.
- ▶ CRRA: $u(z) = z^{1-\gamma}/(1-\gamma)$; $A(z) = \gamma/z$.
- ▶ Logarithmic: $u(z) = \ln z$; $A(z) = 1/z$.
- ▶ Quadratic: $u(z) = cz - \frac{1}{2}z^2$; $A(z) = (c - z)^{-1}$.
- ▶ They all belong to the HARA family:

$$u(z) = \zeta \left(\eta + \frac{z}{\gamma} \right)^{1-\gamma} \implies A(z) = \left(\eta + \frac{z}{\gamma} \right)^{-1}.$$

Your own degree of risk aversion

- ▶ Suppose your current wealth equals 100. There is a fifty-fifty chance of gaining or losing $\alpha\%$ of this wealth.
- ▶ How much are you willing to pay to eliminate this risk?
- ▶ The proportional risk premium π satisfies

$$u(100(1 - \pi)) = \frac{1}{2}u(100(1 - \alpha)) + \frac{1}{2}u(100(1 + \alpha)).$$

- ▶ Suppose CRRA preferences and estimate γ from this equation.

Estimation of relative risk aversion

RRA	$\alpha = 10\%$	$\alpha = 30\%$
$\gamma = 0.5$	$\pi = 0.3\%$	$\pi = 2.3\%$
$\gamma = 1$	$\pi = 0.5\%$	$\pi = 4.6\%$
$\gamma = 4$	$\pi = 2.0\%$	$\pi = 16.0\%$
$\gamma = 10$	$\pi = 4.4\%$	$\pi = 24.4\%$
$\gamma = 40$	$\pi = 8.4\%$	$\pi = 28.7\%$

An application: The social cost of macro risk

- ▶ Homogeneous population.
- ▶ Fair sharing of GDP in each period:

$$\text{consumption}_t = \text{GDP}_t / \text{capita} = c_t.$$

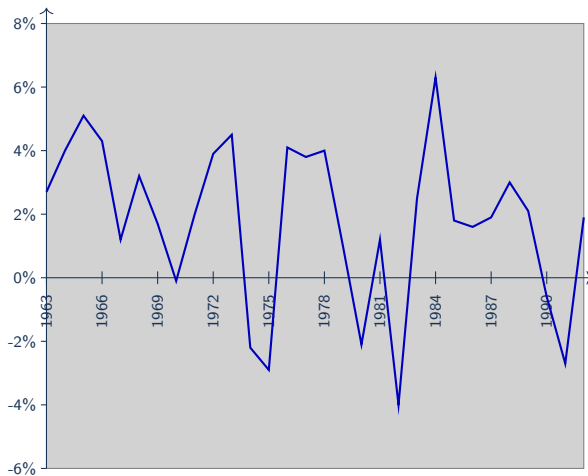
- ▶ $c_{t+1} = c_t(1 + X)$, where X is the growth rate of GDP per capita.
- ▶ The social cost of risk π satisfies

$$\mathbb{E}u(c_{t+1}) = \mathbb{E}u(c_t(1 + X)) = u(c_t(1 + \mathbb{E}X - \pi)).$$

- ▶ Thus π is measured in percentage points of the growth rate.

Uncertain growth of real GDP per capita

Growth of real GDP per capita, USA, 1963–1992



$$\mathbb{E}X = 1.86\% \quad \sigma_X = 2.41\%$$

Calibration

- ▶ Give probability weight $1/30$ to each growth rate observed during the last 30 years.
- ▶ Assume CRRA preferences.
- ▶ $\gamma = 1 \implies \pi = 0.03\%$ loss of growth.
- ▶ $\gamma = 4 \implies \pi = 0.12\%$ loss of growth.
- ▶ The first puzzle of macro-finance.
- ▶ Lucas (1987).

Short bibliography

- ▶ Machina, *Journal of Economic Perspectives*, 1987.
- ▶ de Finetti, 1952.
- ▶ Pratt, *Econometrica*, 1964.
- ▶ Arrow, lecture notes 1963 and book 1970.

The Allais choices

	\$5 million	\$1 million	\$0
a_1	0	1.00	0
a_2	0.10	0.89	0.01
a_3	0.10	0	0.90
a_4	0	0.11	0.89

Many people choose $a_1 \succ a_2$ but $a_3 \succ a_4$.

In-class question

Record both choices before proceeding. Which feature of the first pair is absent from the second?

Why the Allais pattern violates independence

Expected utility implies

$$\begin{aligned}a_1 \succ a_2 &\iff U(1) > .10U(5) + .89U(1) + .01U(0) \\ &\iff .11U(1) > .10U(5) + .01U(0),\end{aligned}$$

$$\begin{aligned}a_3 \succ a_4 &\iff .10U(5) + .90U(0) > .11U(1) + .89U(0) \\ &\iff .10U(5) + .01U(0) > .11U(1).\end{aligned}$$

The two final inequalities contradict one another.

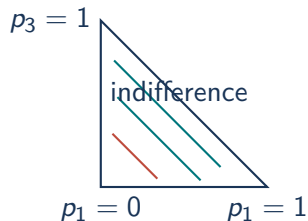
Behavioral reading

The move from certainty to a tiny chance of nothing receives special weight: the **certainty effect**.

The probability triangle

For three ordered outcomes, a lottery is a point in a triangle. Expected-utility indifference curves are straight and parallel because probabilities enter linearly.

- ▶ Parallel lines encode independence.
- ▶ Allais-type choices imply changing slopes.
- ▶ A common empirical pattern is **fanning out**.



Common Consequence Effect

Experiment 1

PROBLEM 1: Choose between

$$A : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 2,400 & \text{with probability } 0.66 \\ 0 & \text{with probability } 0.01 \end{cases} \quad \text{and} \quad B : \begin{cases} 2400 & \text{with probability } 1. \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 0 & \text{with probability } 0.67 \end{cases} \quad \text{and} \quad D : \begin{cases} 2400 & \text{with probability } 0.34 \\ 0 & \text{with probability } 0.66 \end{cases}$$

Common Consequence Effect

Experiment 1

PROBLEM 1: Choose between

$$A : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 2,400 & \text{with probability } 0.66 \\ 0 & \text{with probability } 0.01 \end{cases} \quad \text{and} \quad B : \begin{cases} 2400 & \text{with probability } 1 \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 0 & \text{with probability } 0.67 \end{cases} \quad \text{and} \quad D : \begin{cases} 2400 & \text{with probability } 0.34 \\ 0 & \text{with probability } 0.66 \end{cases}$$

Exercise

Demonstrate that the preferences $B \succ A$ and $C \succ D$ violates expected utility preferences.

Common Ration Effect

Experiment 2

PROBLEM 1: Choose between

$$A : \begin{cases} 4,000 & \text{with probability 0.8} \\ 0 & \text{with probability 0.2} \end{cases} \quad \text{and} \quad B : \begin{cases} 3,000 & \text{with probability 1} \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 4,000 & \text{with probability 0.2} \\ 0 & \text{with probability 0.8} \end{cases} \quad \text{and} \quad D : \begin{cases} 3,000 & \text{with probability 0.25} \\ 0 & \text{with probability 0.75} \end{cases}$$

Exercise

Demonstrate that the preferences $B \succ A$ and $C \succ D$ violates independence axiom.

Isolation Effect

Problem 10 in KT

Consider the following two-stage game. In the first stage, there is a probability of 0.75 to end the game without winning anything, and a probability of 0.25 to move into the second stage. If you reach the second stage you have a choice between

(4,000; 0.80) and (3,000).

Your choice must be made before the game starts, i.e., before the outcome of the first stage is known.

Problem 4 in KT

Choose between

$C : \begin{cases} 4,000 & \text{with probability 0.2} \\ 0 & \text{with probability 0.8} \end{cases}$ and $D : \begin{cases} 3,000 & \text{with probability 0.25} \\ 0 & \text{with probability 0.75} \end{cases}$

Decision Trees for the Isolation Effect

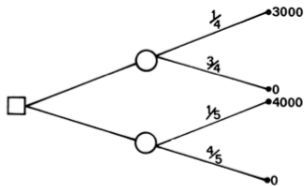
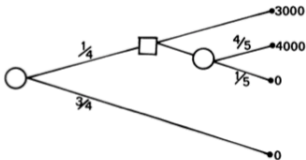


FIGURE 1.—The representation of Problem 4 as a decision tree (standard formulation).



Probabilistic Insurance

Problem 9 in KT

Suppose you consider the possibility of insuring some property against damage, e.g., fire or theft. After examining the risks and the premium you find that you have no clear preference between the options of purchasing insurance or leaving the property uninsured. It is then called to your attention that the insurance company offers a new program called **probabilistic insurance**.

In this program you pay half of the regular premium. In case of damage, there is a 50 percent chance that you pay the other half of the premium and the insurance company covers all the losses; and there is a 50 percent chance that you get back your insurance payment and suffer all the losses. Recall that the premium for full coverage is such that you find this insurance barely worth its cost. Under these circumstances, would you purchase probabilistic insurance?

Another Description of Probabilistic Insurance

Notation: w wealth, L potential loss, π prob of loss, y insurance premium.

Indifference between accepting or rejecting initial policy:

$$\begin{pmatrix} w - L & \pi \\ w & 1 - \pi \end{pmatrix} \sim (w - y)$$

Probabilistic insurance generates the following lottery:

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

Would you choose the probabilistic insurance?

Another Description of Probabilistic Insurance

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Probabilistic insurance generates the following lottery:

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

Would you choose the probabilistic insurance?

Majority (80% of subjects answered NO!)

Risk-Averse EU Agents Prefer Probabilistic Insurance

Notation: w wealth, L potential loss, π prob of loss, y insurance premium.

Indifference between accepting or rejecting initial policy:

$$\begin{pmatrix} w - L & \pi \\ w & 1 - \pi \end{pmatrix} \sim (w - y)$$

Mixing the two, independence implies

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w & (1 - \pi)/2 \\ w - y & (1 - \pi)/2 \end{pmatrix} \sim (w - y)$$

Risk aversion (SOSD monotonicity) then implies

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix} \succ (w - y)$$

Probabilistic Insurance versus Risk of Nonpayment

Risk of nonpayment: α chance of nonpayment with premium y' :

$$\begin{pmatrix} w - L - y' & \alpha\pi \\ w - y' & (1 - \alpha)\pi \\ w - y' & 1 - \pi \end{pmatrix}$$

- Evidence: people tend to require a much more than proportional decrease in premium to tolerate risk of nonpayment: $y' < (1 - \alpha)y$.
- Probabilistic insurance is similar to risk of nonpayment with $\alpha = 0.5$ and $y' = y/2$ (but the two are not quite the same!):

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix} \quad \text{versus} \quad \begin{pmatrix} w - L - y/2 & \pi/2 \\ w - y/2 & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

- A disproportional change in willingness to pay in response to nonpayment risk is difficult to match with EU, but does not yield a direct contradiction of EU like probabilistic insurance.

Violation Patterns

Machina (1987) describes a generalization of the CCE: Let $Q \geq_{\text{FOSD}} R$.

Machina's Hypothesis

PROBLEM 1: Choose between:

$$A : \alpha P + (1 - \alpha)Q \quad \text{or} \quad B : \alpha \delta_x + (1 - \alpha)Q$$

PROBLEM 2: Choose between:

$$C : \alpha P + (1 - \alpha)R \quad \text{or} \quad D : \alpha \delta_x + (1 - \alpha)R$$

- ▶ Independence requires $P \succ \delta_x \iff A \succ B \iff C \succ D$.
- ▶ Machina suggests that violations of independence tend to be of the form $B \succ A$ and $C \succ D$.

What can a generalization change?

The benchmark has two separable ingredients:

$$\underbrace{\sum_i}_{\text{aggregation}} \underbrace{p_i}_{\text{linear probabilities}} \underbrace{U(x_i)}_{\text{outcome utility}} .$$

Alternatives may:

- ▶ transform or rank-weight probabilities;
- ▶ allow nonlinear interaction among lottery components;
- ▶ compare outcomes with foregone alternatives;
- ▶ retain dominance while relaxing independence.

The modeling choice should match the empirical regularity one wants to explain.

Representative non-EU approaches

Approach	Central idea
Weighted utility	Normalize outcome-specific weights; indifference curves need not be parallel.
Moments of utility	Add nonlinear functions of expected utility moments.
Rank-dependent utility	Decision weights depend on cumulative probabilities and outcome ranks.
Quadratic probabilities	Add pairwise probability interactions.
Regret theory	Utility depends partly on what would have happened under the alternative.

No single relaxation is costless: tractability, dominance, dynamic consistency, and descriptive fit can pull in different directions.

Local analysis without committing to one formula

Let $V(P)$ be differentiable in probabilities. The marginal value of shifting probability toward outcome x_i is a generalized probability derivative:

$$U(x_i; P) = \frac{\partial V(P)}{\partial p_i}.$$

Unlike expected utility, this local utility index may depend on the entire lottery P .

- ▶ Under EU: $U(x_i; P) = U(x_i)$.
- ▶ Under general preferences: curvature and slope can vary with location in the probability triangle.
- ▶ This provides a common language for comparing non-EU models.

Regret theory

When two acts are resolved in the same state, the experienced value of outcome x may depend on the forgone outcome y :

$$v(x, y).$$

- ▶ **Regret:** the chosen outcome is worse than the forgone outcome.
- ▶ **Rejoicing:** the chosen outcome is better.
- ▶ Choices depend on the joint state-by-state comparison, not only on each act's marginal outcome distribution.

In-class question

Can two statistically identical lotteries be valued differently when paired against different alternatives?

Subjective expected utility

An act f assigns an outcome $f(s)$ to each state s . Subjective expected utility takes the form

$$V(f) = \sum_{s \in S} \pi(s) U(f(s)).$$

- ▶ U represents tastes over consequences.
- ▶ π represents subjective beliefs.
- ▶ Both are inferred from choices, rather than assumed observable.

Savage's framework constructs probabilities from consistent preferences over acts.

The sure-thing principle

If two acts agree outside event E , their ranking should depend only on what they deliver inside E .

Logic

Consequences common to both acts in states outside E can be cancelled. This is the subjective analogue of independence.

Key supporting ideas include:

- ▶ stable rankings of consequences across non-null events;
- ▶ sufficiently rich events for calibrating likelihood;
- ▶ consistency between event comparisons and act preferences.

A caveat: state-dependent utility

The separation of beliefs and tastes becomes delicate if the same consequence has different utility in different states:

$$V(f) = \sum_s \pi(s) U_s(f(s)).$$

Observed choices can then confound:

- ▶ a state being judged more likely, and
- ▶ its consequences being valued differently.

Additional assumptions or observations are needed to identify beliefs separately from state-dependent preferences.

The Ellsberg urn

An urn contains 30 red balls and 60 balls that are black or yellow in an unknown

		Red	Black	Yellow
mixture.	f_1	\$100	\$0	\$0
	f_2	\$0	\$100	\$0
	f_3	\$100	\$0	\$100
	f_4	\$0	\$100	\$100

Common choices are $f_1 \succ f_2$ and $f_4 \succ f_3$.

In-class question

Choose in each pair. Which bets rely on known rather than unknown probabilities?

Why Ellsberg violates additive beliefs

If preferences use a single additive probability measure π :

$$f_1 \succ f_2 \Rightarrow \pi(R) > \pi(B).$$

But

$$f_4 \succ f_3 \Rightarrow \pi(B) + \pi(Y) > \pi(R) + \pi(Y) \Rightarrow \pi(B) > \pi(R).$$

Contradiction.

Interpretation

People often prefer known risks to poorly specified risks: **ambiguity aversion**. This is distinct from ordinary risk aversion over outcomes.

Two responses to ambiguity

Choquet expected utility

Replace additive probabilities with a capacity. Decision weights can reflect non-additive beliefs and event ambiguity.

Maxmin expected utility

Represent beliefs by a set Π :

$$V(f) = \min_{\pi \in \Pi} \sum_s \pi(s) U(f(s)).$$

Evaluate an act under its least favorable plausible prior.

Both can rationalize Ellsberg-type behavior, but imply different attitudes and comparative statics.

Framing: identical final wealth, different choices

Gain frame

Receive \$1,000, then choose:

- ▶ 50:50 chance of gaining \$1,000 or \$0;
- ▶ a sure gain of \$500.

Most choose the sure gain.

Loss frame

Receive \$2,000, then choose:

- ▶ 50:50 chance of losing \$1,000 or \$0;
- ▶ a sure loss of \$500.

Most choose the gamble.

Both menus imply the same final-wealth distributions: \$1,000/\$2,000 versus \$1,500 for sure.

Description invariance

Standard consequentialist models predict that economically equivalent descriptions produce the same preferences.

- ▶ Frames create reference points.
- ▶ Gains and losses may be coded relative to those reference points.
- ▶ Losses may loom larger than comparable gains.
- ▶ The formulation of a problem can therefore become choice-relevant.

In-class question

When is a frame irrelevant information, and when might it reveal genuinely different consequences to the decision maker?

Procedure invariance and preference reversal

Two gambles are often contrasted:

- ▶ the p -**bet**: high probability of a modest prize;
- ▶ the $\$$ -**bet**: low probability of a large prize.

A recurring pattern is:

direct choice favors the p -bet, pricing assigns a higher value to the $\$$ -bet.

The measured ranking changes with the elicitation procedure.

Possible mechanism

Choice directs attention toward probabilities; pricing makes monetary outcomes especially salient.

What preference reversal challenges

The phenomenon raises three distinct possibilities:

1. preferences are stable, but one elicitation method is biased;
2. preferences are constructed in response to the task;
3. observed responses reflect multiple attributes with procedure-dependent weights.

Thus a complete theory may need to model not only the option set, but also how the decision is described and elicited.

A map of the theories

Model	Key discipline	Main target
Expected utility	Linear probabilities; fixed utility	Risk attitudes and coherent benchmark
Rank-dependent / non-EU	Relax independence	Allais and common-ratio patterns
Regret theory	Statewise comparison with forgone acts	Context and pairwise comparisons
Subjective EU	Additive inferred beliefs	Decisions with unknown states
Choquet / maxmin	Non-additive or multiple beliefs	Ellsberg and ambiguity aversion
Framing / constructed choice	Reference and procedure dependence	Description and response-mode effects

Main lessons

1. Expected utility is powerful because it separates probabilities from utility and yields sharp predictions.
2. Its strongest empirical failures cluster around certainty, small probabilities, ambiguity, and context.
3. Different paradoxes challenge different pieces of the benchmark; they do not all call for the same repair.
4. A useful model balances descriptive fit, normative appeal, identification, and tractability.

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